SHADOWFOX DATA ANALYST INTERNSHIP
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Beginner Superstore analysis

Rebuild, validation and findings report
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EXECUTIVE SUMMARY

The Level 1 project cleans the Sample Superstore export and builds a live Excel dashboard from 9,993 order lines, 5,009 orders, 793 customers and 1,862 product IDs. The complete pipeline now runs in the documented order and passes a regression validator. The resulting workbook contains KPI definitions, monthly trend and seasonality, product and discount analysis, a dashboard, and an auditable cleaning log.

Metric	Definition	Result
Sales	Sum of cleaned line sales	$2,296,919.49
Profit	Sum of cleaned line profit	$286,409.08
Profit margin	Profit divided by sales	12.5%
Orders	Distinct order IDs	5,009
Return rate	Returned orders divided by orders	5.9%

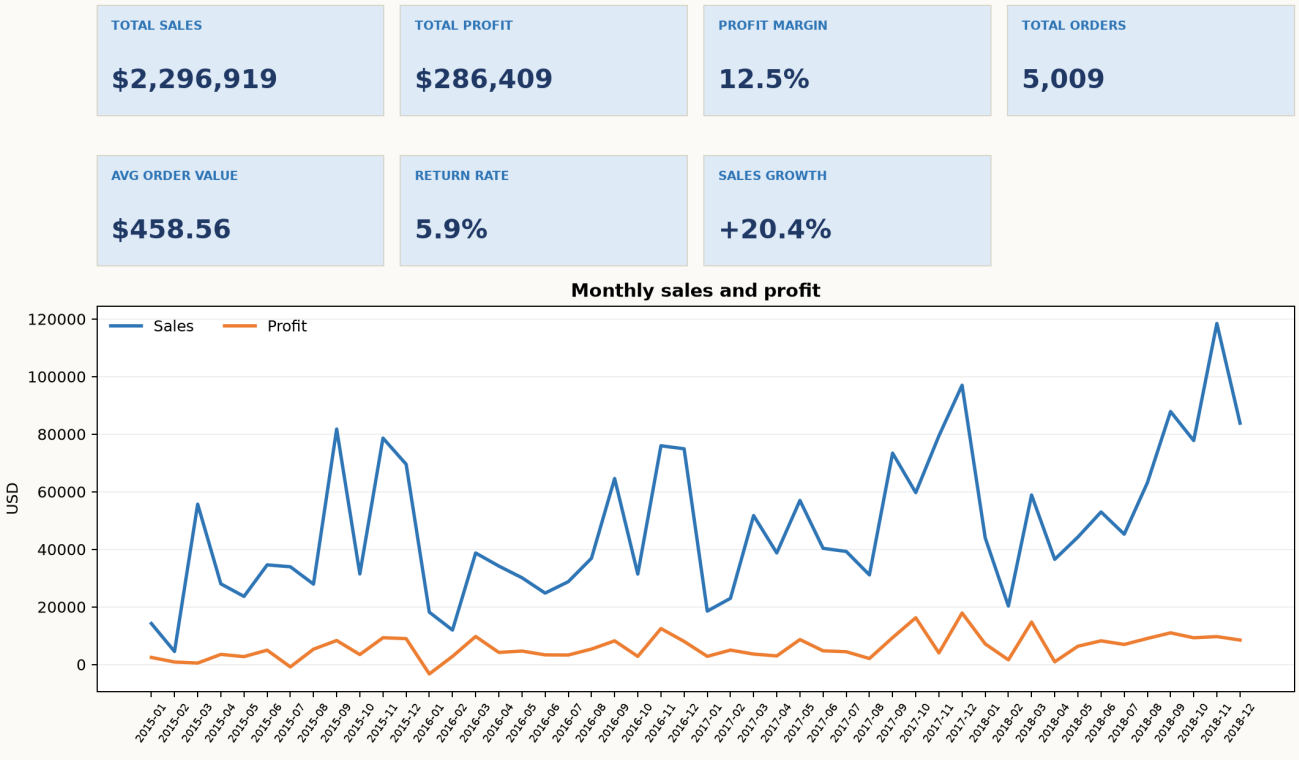


Figure 1: Dashboard-style overview reconstructed from the cleaned data. The seven KPI cards and the monthly sales/profit curve correspond to the main Excel Dashboard sheet.

WORKBOOK DASHBOARD AND VISUAL EVIDENCE

The Excel workbook is not only a table of results: it is a formula-driven dashboard. The report therefore includes the same visual story in a portable PDF form. Values in these figures are recalculated from `data/clean/superstore_clean.csv`; the workbook remains the interactive version with formulas, charts and filters.

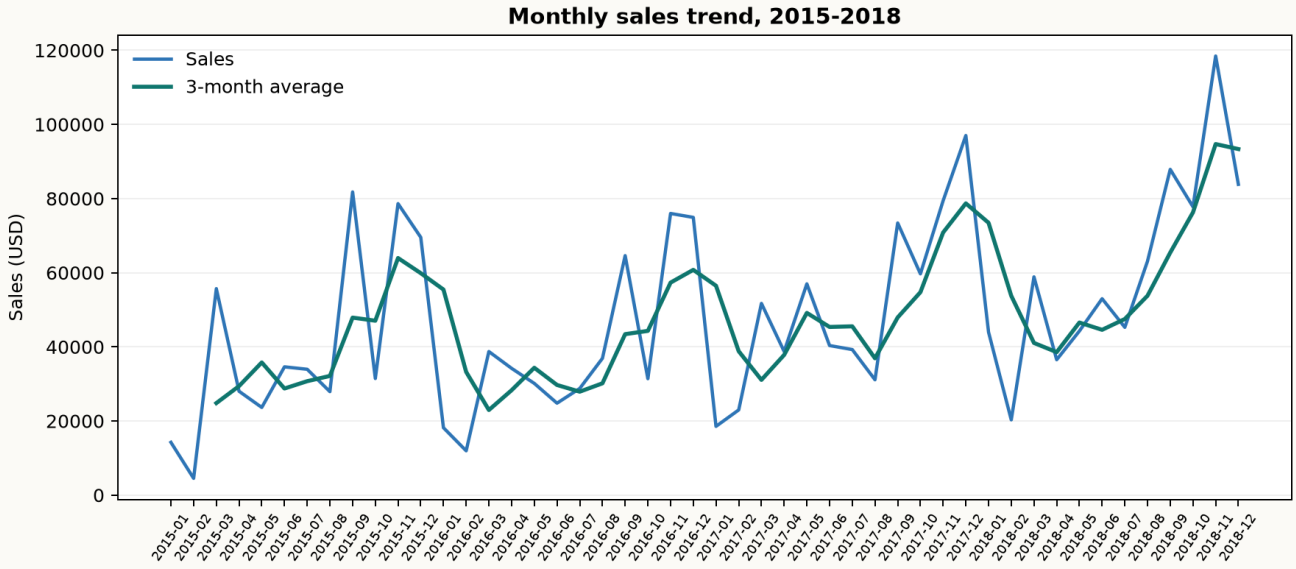


Figure 2: Monthly sales curve and three-month moving average. The curve shows recurring year-end peaks and a clear increase in the later years.

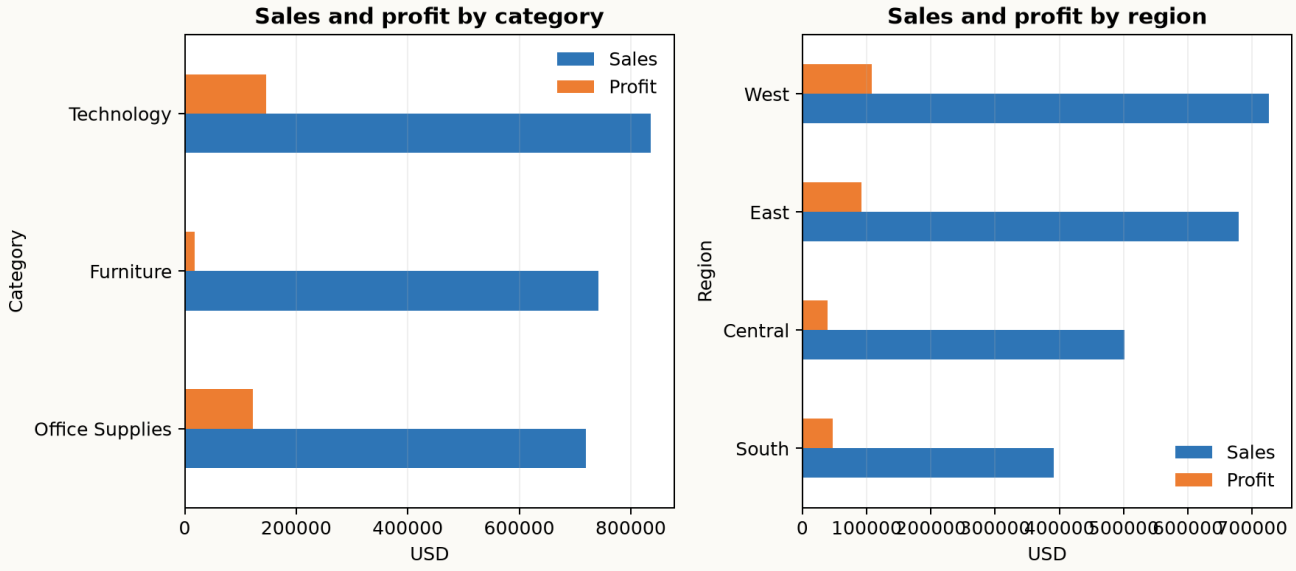


Figure 3: Sales and profit by category and region. Furniture is much less profitable than Office Supplies and Technology, while the West and East carry the largest sales totals.

WHAT WAS FIXED

- The cleaner now stops with an explicit error when any integrity check fails, instead of writing "FAILED" to the log and continuing.
- "First Line of Order" now uses a running `COUNTIF`, so order counts remain correct even if an input file is not grouped by order.
- A new `validate_project.py` script checks row counts, totals, data constraints, required sheets, formulas and dashboard structure.
- The README source path was corrected, the screenshot placeholder was replaced with usable workbook instructions, and recommendations were added.

REPRODUCIBLE WORKFLOW

```
cd "d:\ShadowFox Intership\shadowfox-data-analyst\01-beginner"
python scripts\clean_superstore.py
python scripts\build_workbook.py
python scripts\add_analysis.py
python scripts\add_dashboard.py
python scripts\validate_project.py
```

The raw CSV is preserved. The first script splits the combined Orders, People and Returns export, converts types, fills the 11 missing Burlington ZIP codes, removes one duplicate line and records every action in the cleaning log. The three workbook-building scripts are intentionally ordered because later scripts add sheets and formulas to the workbook produced by earlier scripts.

BUSINESS FINDINGS

1. Deep discounting is the clearest margin warning. Lines above 20% discount are loss-making in aggregate, so discount approval should be reviewed by sub-category rather than applied broadly.
2. Furniture is the weakest category. Tables and Bookcases are the main sub-category loss areas; Machines also combines substantial sales with a low margin and a high average discount.
3. Demand is seasonal. September, November and December contribute about 43% of sales, while January and February contribute about 7%. The workbook compares calendar months across all four years.
4. Product concentration matters. A small number of products account for very large gains or losses, making product-level review more useful than product-ID-only grouping because 32 IDs map to multiple names.
5. Returns are material but not dominant: 5.9% of orders and 7.9% of sales are associated with returned orders. Headline figures remain gross because the source does not state whether profit was reversed.

CATEGORY, PRODUCT AND MARGIN DETAIL

WORKBOOK MAP

Sheet	Purpose
Dashboard	KPI cards and charts for the executive view.
Monthly Trend	48-month sales/profit table, moving average and seasonality grid.
Product Analysis	Sub-category performance and top/bottom product rankings.
Discount Analysis	Band, exact-rate, category, monthly and correlation analysis.
KPI Definitions	Live KPI formulas, definitions and supporting category/region tables.
Data	Clean order lines plus formula-derived year, quarter, ship days, returns and discount band.
Cleaning Log	Auditable record of every split, conversion, correction and validation check.

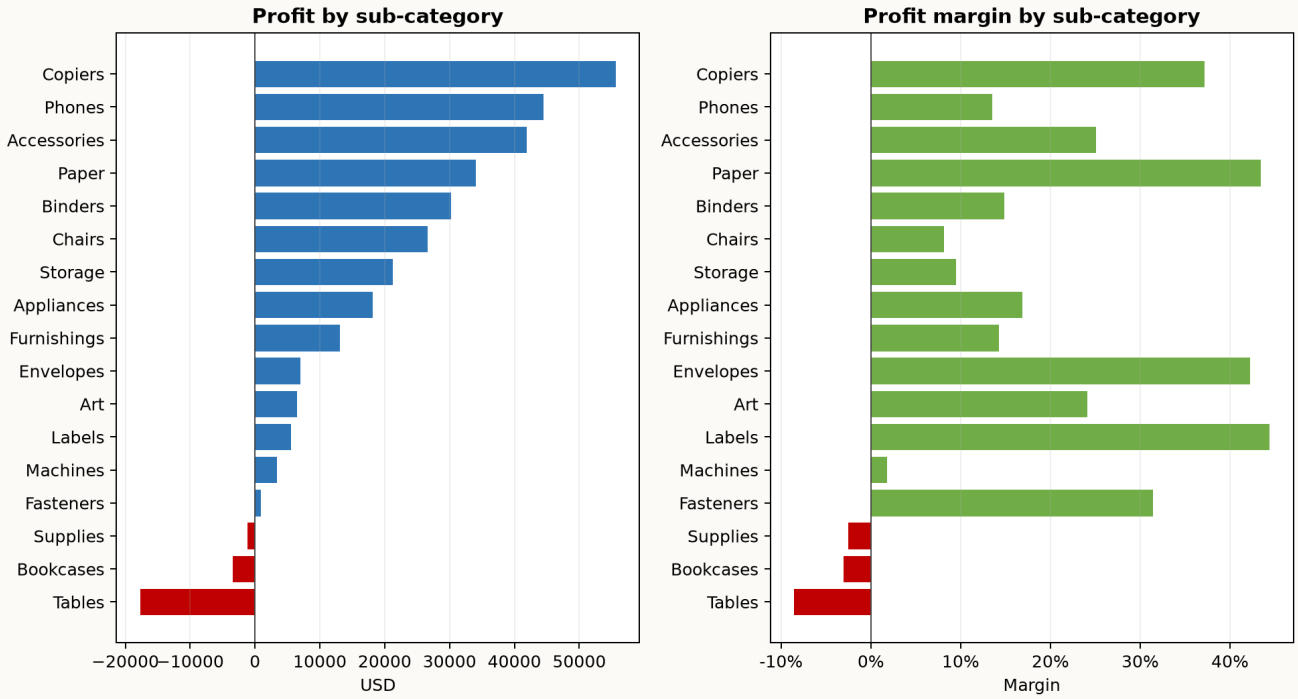


Figure 4: Sub-category performance. Red bars identify negative profit or negative margin; this makes Tables and Bookcases immediately visible as priority areas.



Figure 5: Top and bottom ten products by aggregate profit. Product names are used rather than Product IDs because the source reuses 32 IDs for different names.

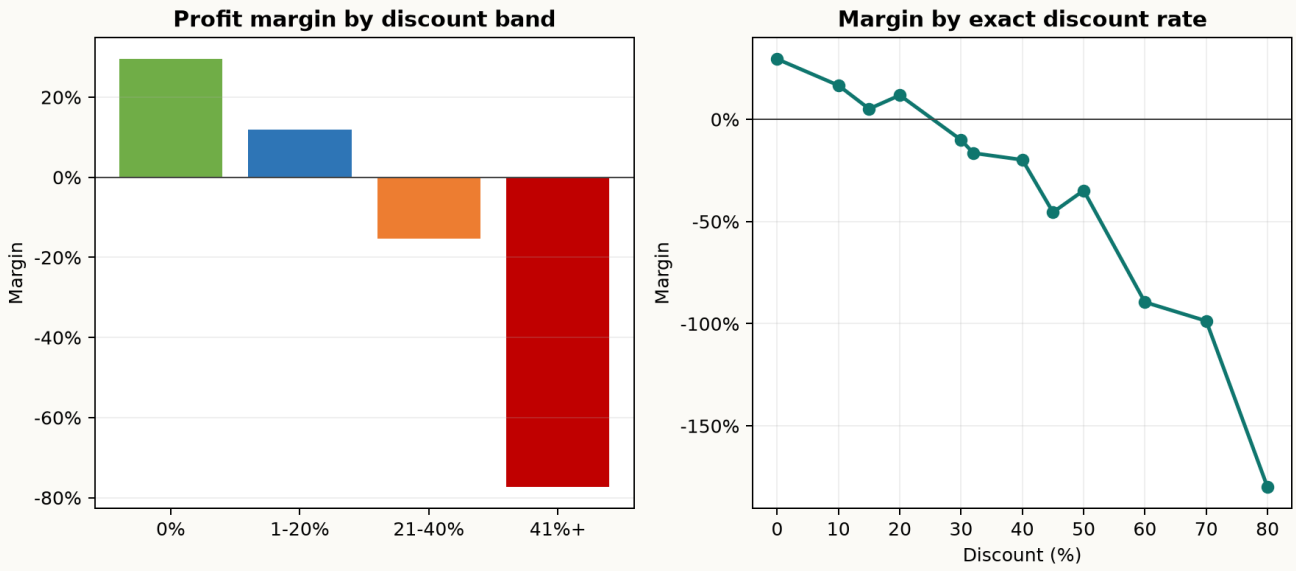


Figure 6: Discount analysis. The left chart groups lines into the workbook's four discount bands; the right chart shows the relationship at each exact discount rate.

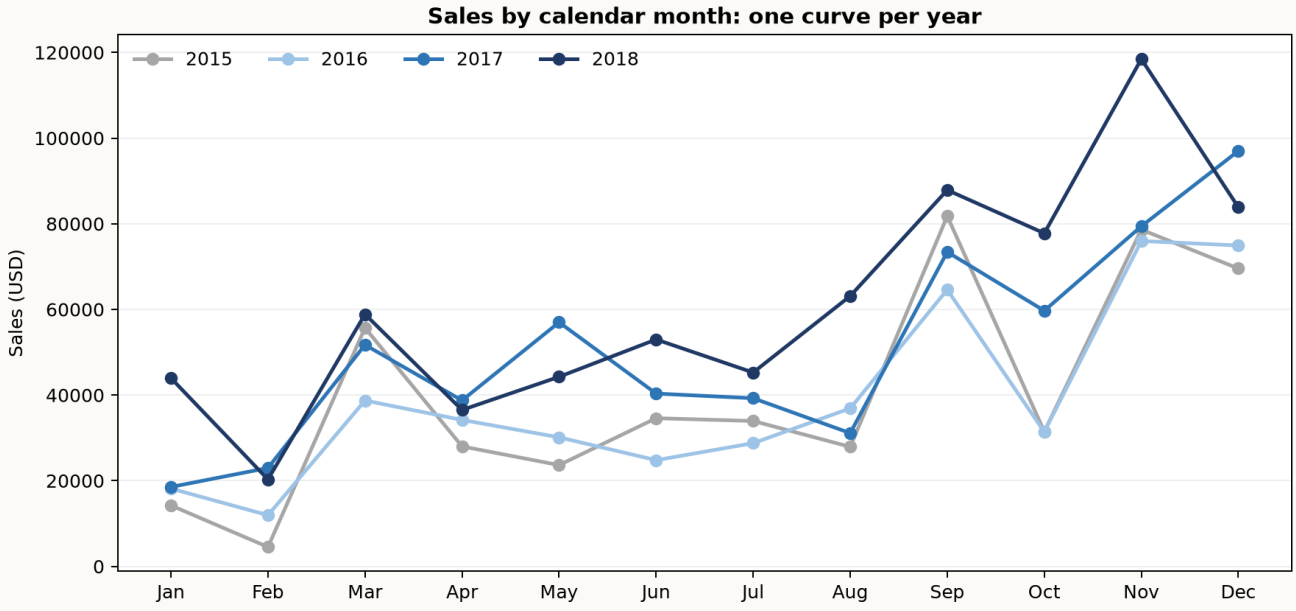


Figure 7: Seasonality curve from the Monthly Trend sheet. Each line is one year, allowing the same calendar month to be compared across years.

RECOMMENDATIONS

1. Introduce a review or minimum-margin guardrail for discounts above 20%, beginning with Tables, Bookcases and Machines.
2. Protect peak-season demand with availability, bundles and cross-sells before adding broad discounts.
3. Join returns, fulfillment cost and customer data to test whether the worst products and returned orders are structurally unprofitable.

LIMITATIONS

The discount results are associations, not causal estimates. The data covers one retailer from 2015–2018 and does not contain cost detail, marketing exposure or customer contact history. Return profit treatment is unknown, and the Burlington ZIP fill is an outside factual assumption. Excel formula cells require recalculation in Excel or LibreOffice after generation.

NOTES

- **What changed in the prompt:** Added explicit validation and made the README operational rather than placeholder-based.
- **What improved in the output:** The project is reproducible, the workbook structure is checked, and the recommendations connect directly to measured findings.
- **What still failed:** No causal experiment can be inferred from this observational dataset, and slicers remain a manual PivotTable step.
- **What I would try next:** Add a controlled discount test and join margin, cost and return outcomes before changing commercial policy.

DELIVERABLES

- Live Excel workbook
- Regression validator
- This LaTeX source